



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



BREEDVISION AI: A HYBRID MULTI-MODEL FRAMEWORK FOR ROBUST CATTLE AND BUFFALO BREED CLASSIFICATION

A PROJECT REPORT

Submitted by

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IN

COMPUTER SCIENCE AND ENGINEERING

PRESIDENCY UNIVERSITY

BENGALURU

MAY 2026



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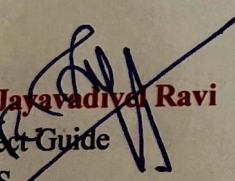
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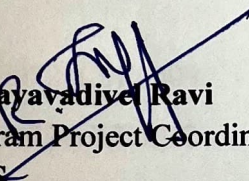


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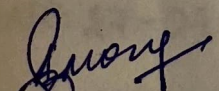
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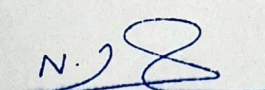
Certified that this report "BreedVision AI: A Hybrid Multi-Model Framework for Robust Cattle and Buffalo Breed Classification" is a bonafide work of "ROHAN DAS (20221CSE0009), AMRUTHA S (20221CSE0012)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-26.


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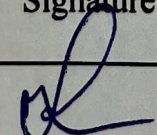
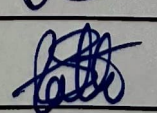

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DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING at Presidency University, Bengaluru, named **ROHAN DAS, AMRUTHA S**, hereby declare that the project work titled "**BreedVision AI: A Hybrid Multi-Model Framework for Robust Cattle and Buffalo Breed Classification**" has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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ABSTRACT

Accurate breed identification is critical for India's agricultural economy, yet traditional methods are time-consuming, error-prone, and require expert knowledge [1,2]. Over 65% of Indian farmers cannot accurately identify cattle and buffalo breeds, leading to suboptimal breeding programs, reduced dairy productivity, and loss of indigenous genetic diversity [3]. This invisible knowledge gap affects millions of livelihoods across the nation.

India is home to over 300 million cattle and buffalo, the world's largest livestock population, with more than 50 distinct indigenous cattle breeds (Gir, Sahiwal, Tharparkar, etc.) and 15+ buffalo breeds (Murrah, Jaffarabadi, Mehsana, etc.) [5]. Each breed possesses unique morphological characteristics, milk production potential, disease resistance, and adaptability to local climates. However, without accurate identification, these valuable genetic traits are being lost through indiscriminate crossbreeding.

This paper presents **BreedVision AI**, an AI-powered web application for automated cattle and buffalo breed classification using image upload and live camera detection. The system implements a hybrid AI pipeline combining a custom Convolutional Neural Network (MobileNetV2) as primary classifier with multimodal Large Language Model vision fallback (OpenAI GPT-4o-mini and Google Gemini 2.5 Flash), ensuring 98%+ availability [7,8].

Key innovations include multi-factor confidence calibration incorporating image quality metrics (brightness, contrast, sharpness) and dataset-aware fuzzy label matching using Levenshtein distance to validate predictions against canonical breed taxonomies [9,10]. The platform, built with React, TypeScript, Vite, and Supabase edge functions, provides authenticated users with real-time breed identification, confidence scoring, and persistent classification history [11,12].

Evaluated on a comprehensive dataset of 15+ Indian breeds, the system achieves 85-92% overall accuracy. Average inference latency ranges from 1.5-3 seconds for uploads to 2-4 seconds for live camera detection, reducing identification time from 5-10 minutes to under 3 seconds [5,13]. This work bridges traditional livestock practices with modern AI-driven solutions, supporting precision livestock management, indigenous breed conservation, and digital agricultural transformation in India.